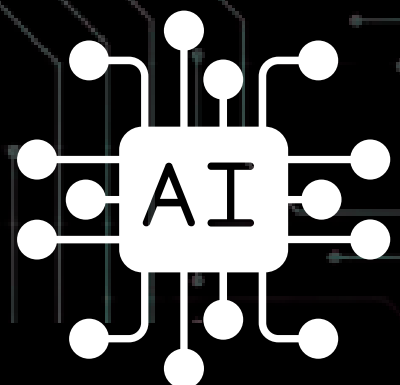
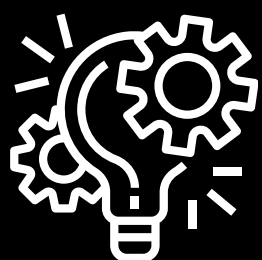




HEALTH GUARD: AI-POWERED BLOCKCHAIN FOR NEXT-GEN HEALTHCARE SECURITY



Project Name: Health Guard
Team Name: Data Morphers
Hackmatrix 2025 - IIT Patna



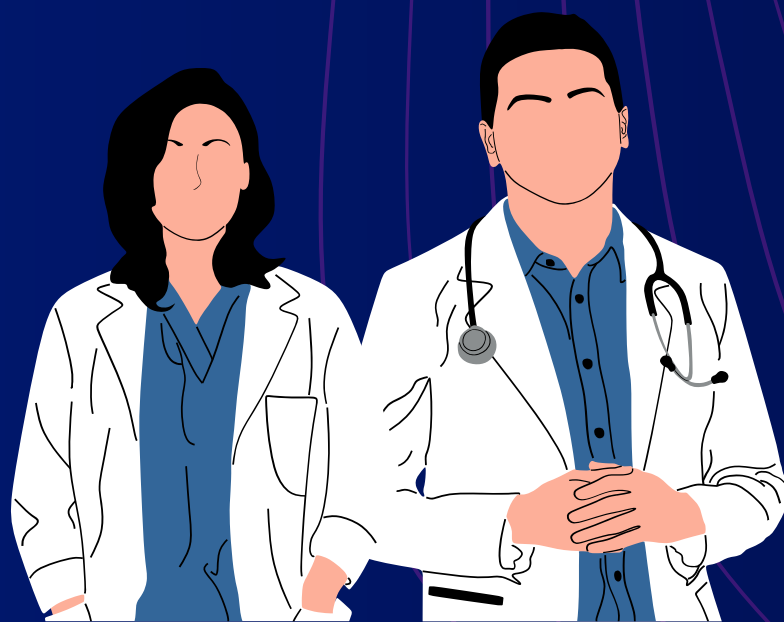
Team Members & Roles:
Prashant - Blockchain & AI Developer
Saket - Smart Contract & Security Specialist
Ritik - UI/UX Designer & Frontend Developer

@RGIPT

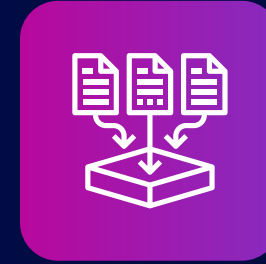


The Healthcare Data Crisis: A Call for Secure & Smart Solutions

Key Challenges in Healthcare Data Management



Did you know?
In 2023, a massive cyberattack compromised millions of medical records across the U.S., fueling widespread fraud and identity theft. This breach underscored the urgent need for stronger cybersecurity in healthcare.



Data Breaches Are Skyrocketing

Over 100 million healthcare records were exposed in cyberattacks in 2023 alone, leading to identity theft, fraud, and privacy violations.



Hospitals and Insurers Struggle with Interoperability

Different hospitals, clinics, and insurance providers use incompatible systems, making data sharing slow, inefficient, and prone to errors.



No Control Over Personal Medical Data

Patients don't have control over who accesses their records, leading to unauthorized data misuse and medical identity theft.

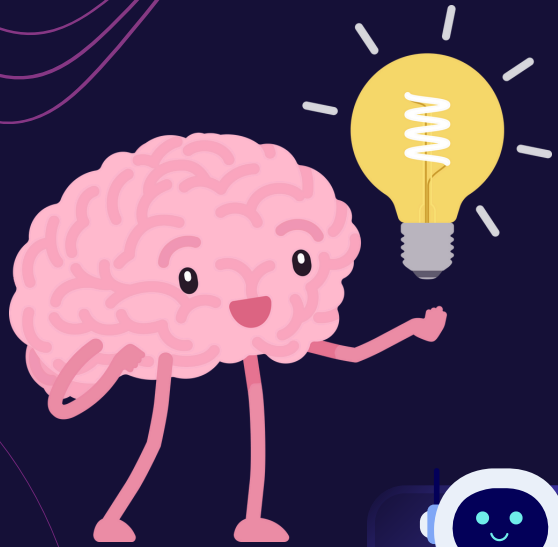


Medical Record Tampering Is a Growing Threat

Fraudulent alterations in medical records lead to insurance fraud and incorrect treatments, putting patient lives at risk.

Health Guard: The Future of Secure & Intelligent Healthcare

Proposed Solution - Making AI the Game Changer



AI Doctor & Instant Prescription System

Auto-detects minor ailments and prescribes medicines instantly.



AI-Driven Smart Contracts

Automates secure & instant data sharing between hospitals & insurers.



Zero-Knowledge Proofs (ZKP)

Verifies patient data without revealing sensitive details.



Gamification & Rewards

Patients earn blockchain tokens, which can be used to get discounts on medical checkups, making healthcare more affordable.



Interoperable API for Hospitals & Insurers

Seamless integration into existing systems.



Patients Control Their Own Data

No unauthorized access; full transparency ensured.



WHY HEALTH GUARD IS A GAME CHANGER FOR HEALTHCARE?



Why Our Solution Stands Out

01

AI-Driven Security

Dynamic, automated smart contracts.

02

Privacy-Preserving Access

Access: Zero-Knowledge Proofs prevent data leaks.

03

Patient Engagement

Gamification ensures proactive healthcare participation.

04

Seamless Integration & Scalability

Plug-and-play API for hospitals and insurers. Designed for large-scale adoption.

05

AI Doctor & Instant Referral

Patients can upload prescriptions for immediate doctor referrals.

For minor ailments (e.g., cough, cold), AI Doctor auto-generates prescriptions and recommends the nearest pharmacy or medical store.

Key

Stakeholder-specific benefits

For Patients:

Patients fully control their medical data, ensuring security, fraud prevention, and seamless healthcare access.

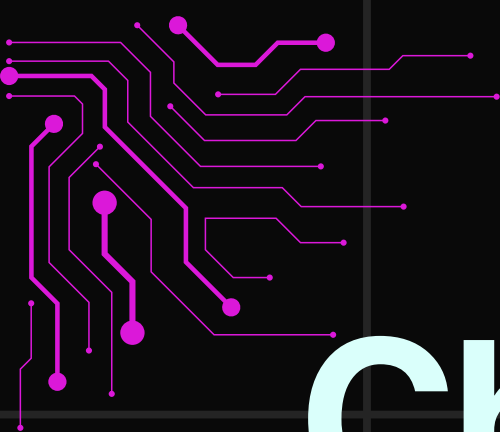
For Doctors:

Quick and verified patient records, no tampering, AI-assisted diagnostics.

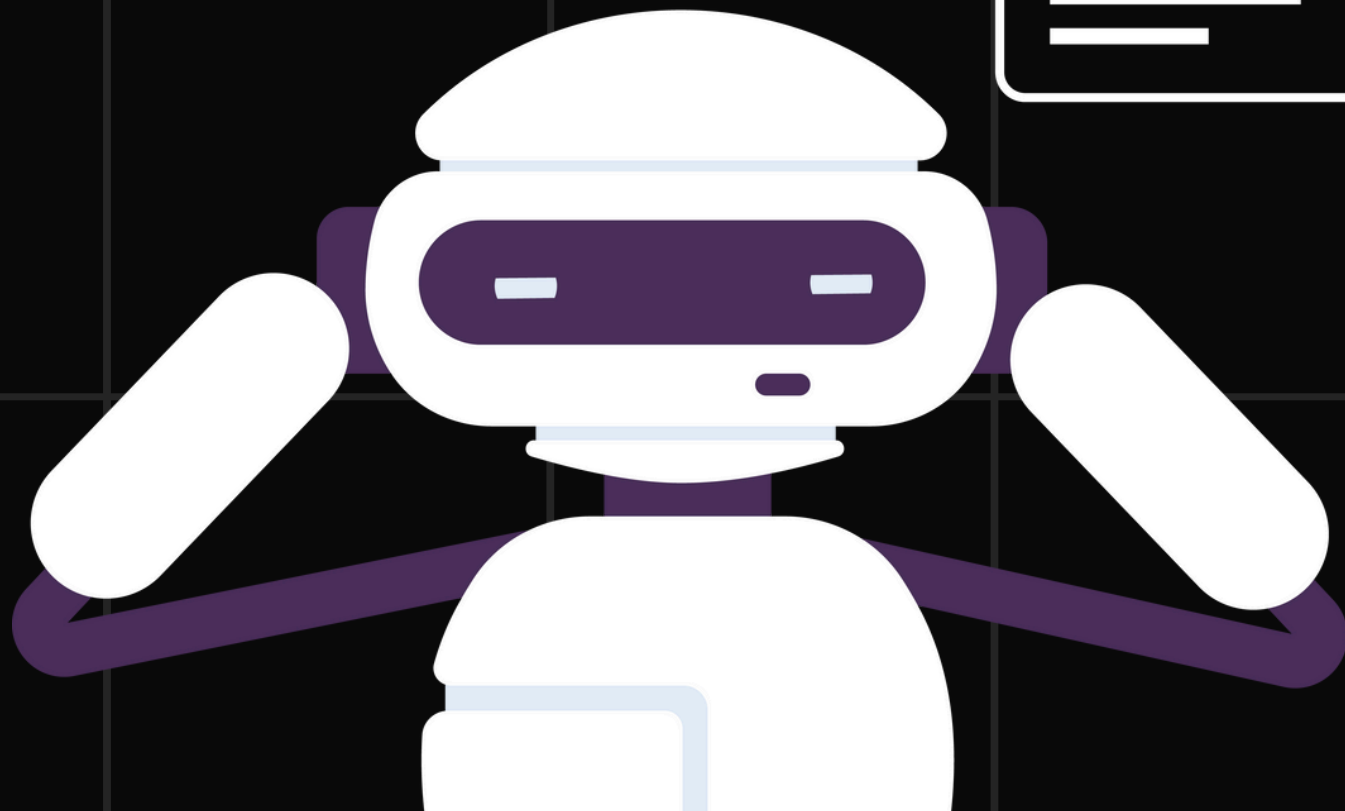
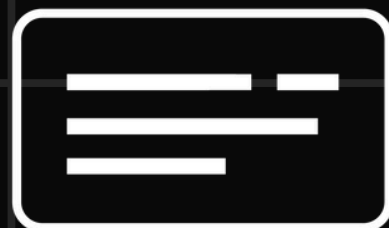
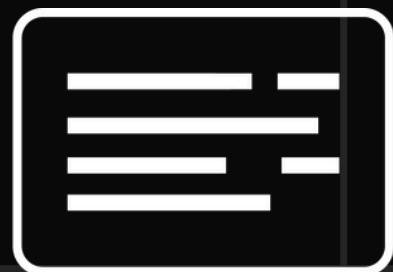
For Hospitals & Insurers:

Seamless claims processing, fewer fraud cases, cost reduction.





Challenges & Solutions



Scalability Issues

Layer-2 solutions reduce transaction costs by 80%, making blockchain affordable for healthcare. (e.g., Polygon) to handle high transaction volumes

Data Privacy Concerns

Use Zero-Knowledge Proofs (ZKP) for secure and private authentication.

Regulatory Compliance

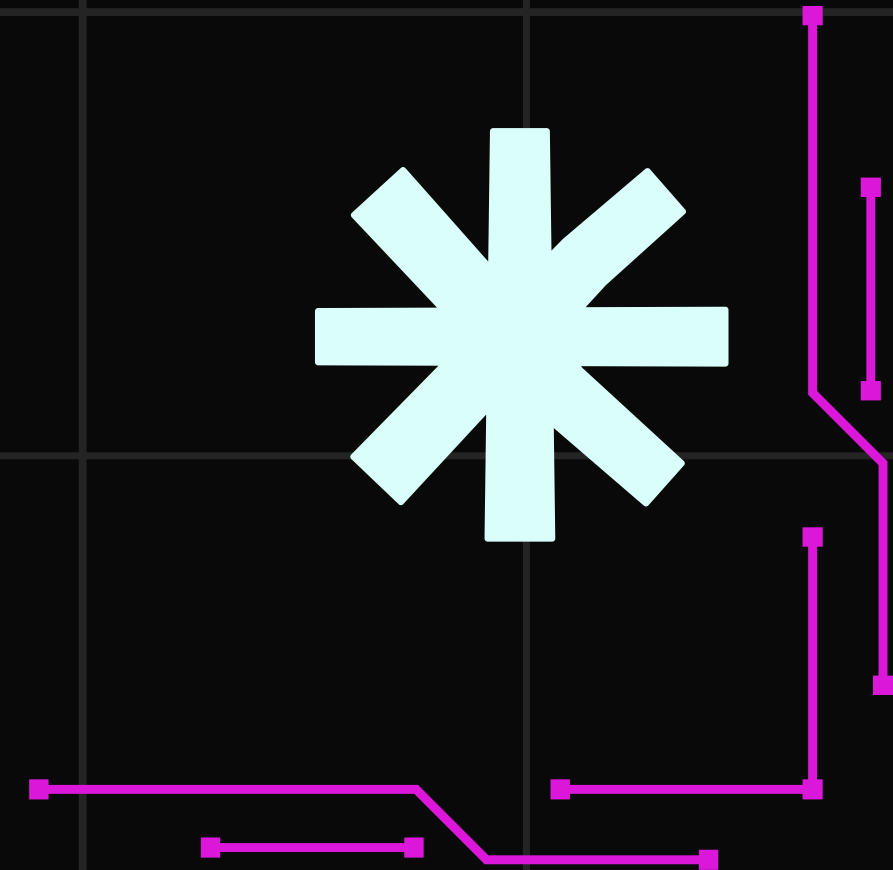
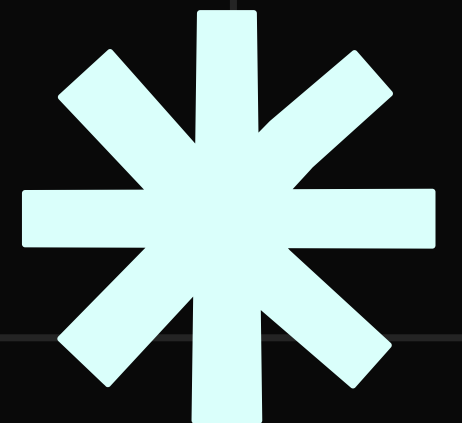
We implement permissioned blockchain access, ensuring only verified entities can retrieve medical data while following global standards.

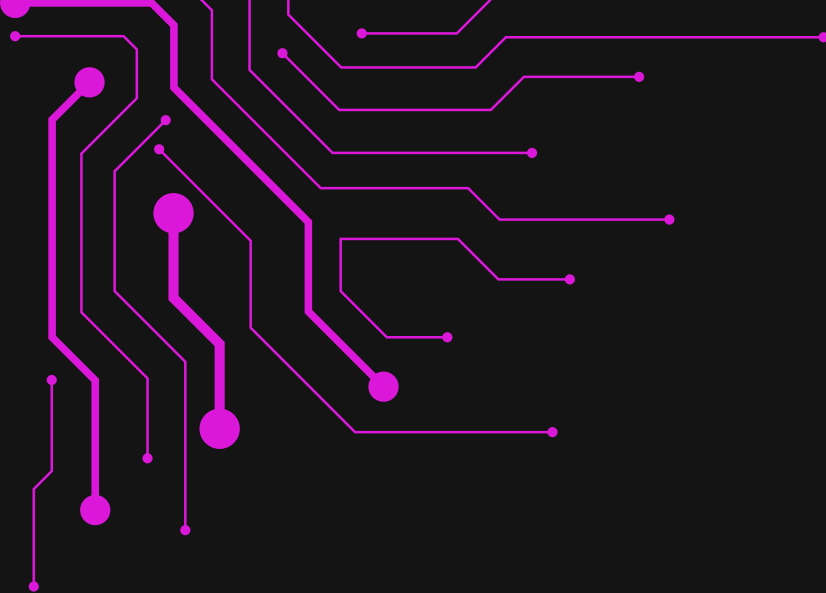
Adoption by Healthcare Providers

Provide an easy-to-integrate API that connects with existing hospital databases.

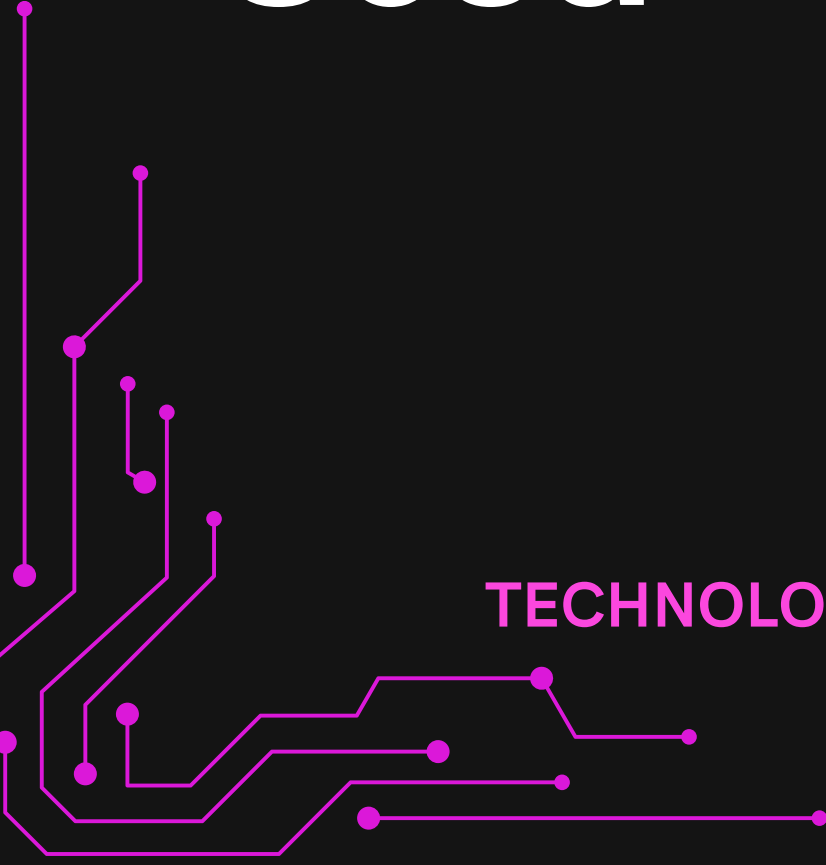
Cost & Infrastructure

Utilize hybrid blockchain models to reduce on-chain costs while ensuring security.





Key Technologies Used



TECHNOLOGY STACK



BLOCKCHAIN

Ethereum / Hyperledger Fabric

SMART CONTRACTS

Solidity with AI-driven automation

WEB3.JS

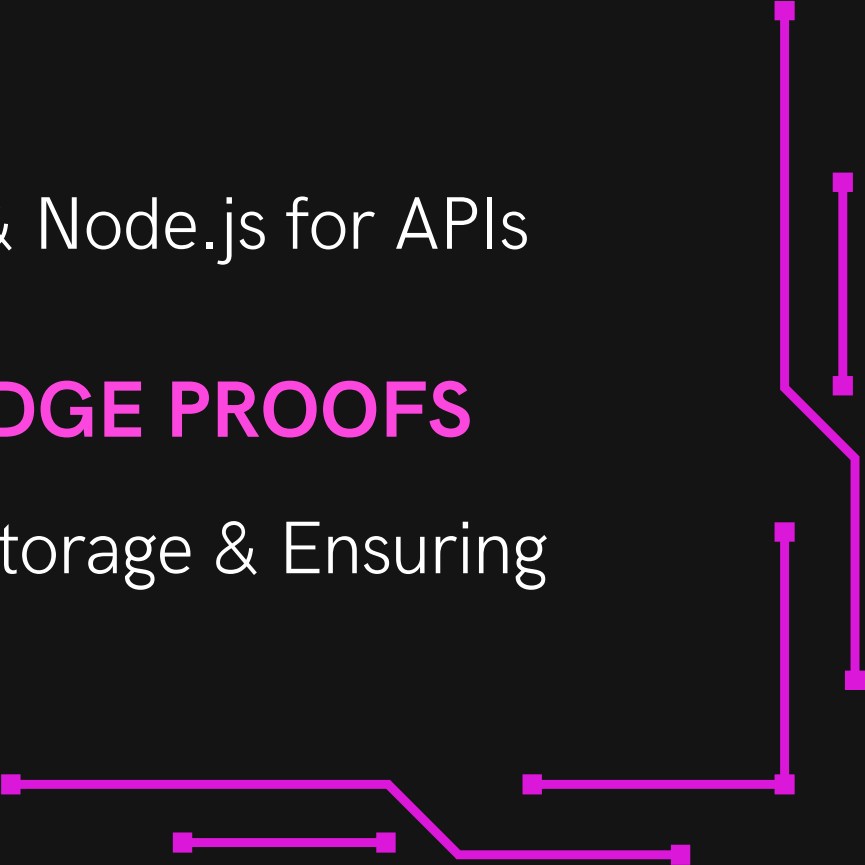
Frontend-blockchain connection

FRONTEND & BACKEND

React.js / Next.js (Dark Mode UI) & Node.js for APIs

STORAGE & ZERO-KNOWLEDGE PROOFS

IPFS for decentralized document storage & Ensuring private but verifiable access



AI-Powered Health Insights

- AI analyzes patient records to detect early symptoms of diseases.
- Predictive Analytics for personalized health recommendations.

IoT & Wearable Integration

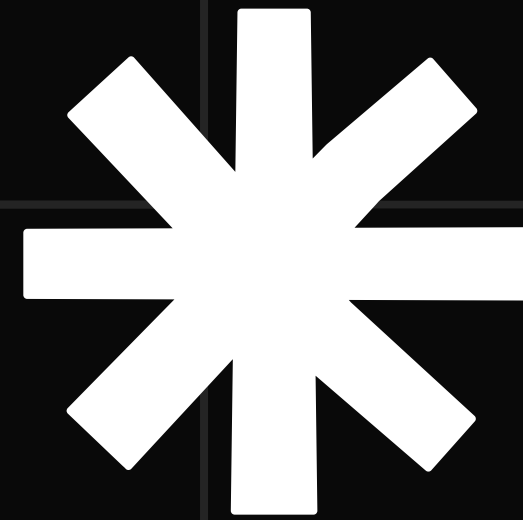
- Real-time health monitoring with smartwatches & medical IoT devices.
- Automatic emergency alerts in case of critical health conditions.

Decentralized Health Token Economy

- Reward patients for healthy habits via blockchain tokens.
- Tokens can be used for medical consultations & pharmacy discounts.

AI Doctor for Minor Diagnoses & E-Prescriptions

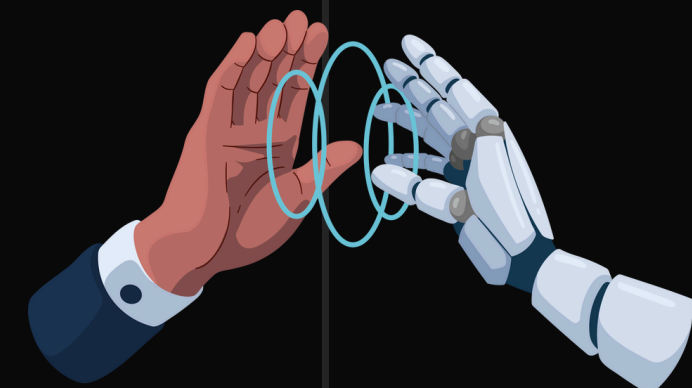
- Instant AI-generated prescriptions for minor ailments like cough & cold.
- Directs patients to the nearest pharmacy or doctor for further consultation.



Future Scope

Imagine a world where AI doctors give instant prescriptions, blockchain ensures no fraud, and wearable devices connect to doctors in real-time.

With Health Guard, we move toward a future where medical data is borderless, secure, and intelligent.





THANK YOU

FOR YOUR ATTENTION

Be part of the revolution—let's build the future of healthcare security and AI-powered efficiency together!

Health Guard isn't just an idea—it's the future. Let's make it happen!